

Some important lessons for radiomen are carried by this



Light Sensitive . . .

ELECTRONIC BEAST

By EDMUND C. BERKELEY

ON the cover of this issue of RADIO-ELECTRONICS is a picture of a small robot which has four sensing organs, three acting organs, and a small electronic and relay brain. His name is Squee, the Robot Squirrel. What he does is roll along the floor, hunt for "nuts," pick up a "nut" in his scoop, take it over to his "nest," leave it there, and then hunt for more nuts.

Although Squee is not a very clever robot, he does have a small amount of memory and of reasoning ability, and he is a close cousin of his predecessor, Simon, the Midget Electric Brain. Simon was the main subject of a series of thirteen articles in RADIO-ELECTRONICS from October, 1950, to October, 1951, by Robert A. Jensen and this author.

There are a number of interesting things about Squee, himself, but the most important of them is that he is in many ways a good illustration of a powerful new method for the design of

circuits for mechanical brains and robots. This method is the *algebra of logic*, also called *Boolean algebra*. The engineers at Northrop Aircraft Co. in California, who designed the electric brain Maddida, say they have given up drawing circuit diagrams in many places because Boolean algebra does a better job.

In this series of two articles the main emphasis will be on Boolean algebra: what it is, how you can calculate with it, and how it can be used in practice. The secondary emphasis will be on Squee. But first a few more words about Squee.

Why did Edmund C. Berkeley and Associates build Squee? Well, last year Bob Jensen and I read some articles about a mechanical tortoise made in Bristol, England, by Dr. W. Grey Walter at the Burden Neurological Institute. We said to ourselves, "Let's make a robot like that—but have him do something a little more clever." So we came up with the idea of a squirrel gathering nuts, and decided to make a robot squirrel.

Squee was constructed mainly through the efforts of three men—Robert A. Jensen (until he re-entered the Air Force in June, 1951), William Szabo, and Jack Koff. Bob Jensen made the skeleton, a framework holding a front wheel for driving, a pivoted column

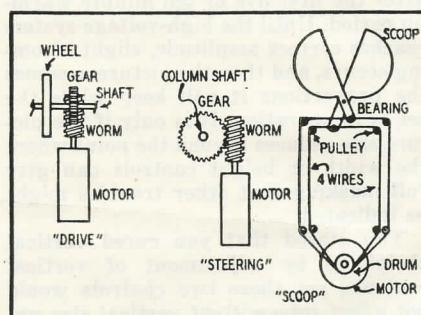


Fig. 2—These are the squirrel's motor organs.

holding and steering the wheel, and two rear wheels rolling free. He mounted tubes, relays, and batteries. By October, 1950, the machine was responsive to light (but only to one kind of light). It still had no "hands." In March, 1951, we made a commitment to exhibit Simon and Squee at the Minnesota State Fair, August 25 to September 3, under the auspices of The Dayton Co., a large department store of Minneapolis. The project that had started as fun became good business. We undertook in earnest the work of making the machine sensitive to a second kind of light, and to give it the needed hands. The scheme for the scoop and the nest light was the main contribution of Szabo, and the final completion, testing, and modi-

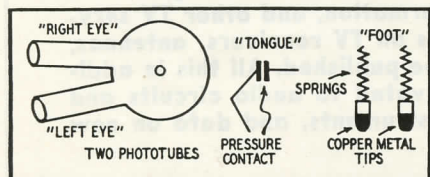


Fig. 1—The electronic sensory organs of Squee.

fication of the machine was due to Jack Koff. Squee was exhibited ten hours a day for ten days in Minneapolis, and probably 50,000 people saw him.

The design of Squee

The physical design of Squee is based on hitching together the sensing organs (see Fig. 1) and the acting organs (see Fig. 2) with appropriate hardware. The logical design of Squee is based on the Boolean algebra relating the conditions expressed by the sensing organs and the conditions expressed by the acting organs.

The first two sensing organs are the *right eye* and the *left eye*, the two photocells. They enable Squee to scan the surrounding environment; as Squee turns his steering column, the photocells look in one direction after another for nuts or nest. A nut (currently a golf ball) is lighted from above by a steady light, a d.c. light. The nest (currently a 12 x 18-inch sheet of aluminum) is lighted by a 60-cycle a.c. gas-filled lamp giving 120 flickers a second.

In Squee the physical circuit using electronic tubes connected to each photocell reports at any time three logical conditions. These are: darkness; a.c. light; and either d.c. or a.c. light or both a.c. and d.c. light. The possible logical reports from the circuit, for each phototube are:

Condition Darkness A.C. Any light

No.	Report	Report	Report
1	1	0	0
2	0	1	1
3	0	0	1

Here the 1 designates "yes" or "reported" or "on," and the 0 denotes "no" or "not reported" or "off." Notice particularly that this circuit, which we called the Amplifying Circuit, was unable to report "d.c. and not a.c."; there will be a lot to say about this point later.

The third sensing organ of Squee is a contact-reporting switch taken from a vending machine, and installed at the base of the scoop. We called this the "tongue." When the nut (ball) entered the scoop, it would roll against this contact switch and close a relay, thus telling Squee that it had taken hold of a nut.

The fourth sensing organ of Squee is a "foot," consisting of two copper tips mounted on springs, which trail along the floor. If and when both of them touch a metal plate (the "nest"), a relay is closed, and Squee "knows" that it has found the nest.

The possible logical reports from these two sensing organs are:

Tongue Report	Foot Report
1	1
0	0

We come now to the acting organs. After a lot of pondering over various ways of giving energy to the acting organs, and the problems of clutches,

we finally decided on the simplest, though crudest method: We hitched a separate motor to each part that had to be moved, and we provided that it could be de-energized, run forward or backward.

For the drive wheel, we mounted a gear on the drive shaft, and turned that gear with a worm wheel mounted on the shaft of the drive motor. For steering, we mounted a gear on the column shaft, and turned the gear with a worm wheel mounted on the shaft of the steering motor. In the case of the scoop, we had a problem. There was room to put a motor at the bottom rear of the chassis. But the scoop was at the front of the chassis, even ahead of the column, and it had to be opened and closed like two cupped hands held together at the wrists. So we ran pulley lines made of light, flexible wire string, from the base of the scoop to the drum mounted on the shaft of the scoop motor; and we adjusted the amount of

turning of the motor by means of limit switches, so that there would be two positions and the scoop would be either open or closed.

The possible logical reports about each of these three acting organs is:

Condition	Motor On	Motor Forward	Motor Backward
1	0	1 or 0	0 or 1
2	1	1	0
3	1	0	1

We have now reduced the sensing (or input) of Squee to a set of yes's and no's, or 1's and 0's. We have reduced the acting (or output) of Squee to a set of yes's and no's, or 1's and 0's. We now have left the problem of hitching the input and the output together, so as to express the desired behavior of Squee.

Ordinarily, up to this time, this kind of problem has been solved by the practical method of drawing circuits on paper, using prior rule-of-thumb experience with that method. But there

CHART 1—THE IDEAS OF BOOLEAN ALGEBRA

ELEMENTARY ALGEBRA QUESTION BOOLEAN ALGEBRA

1. What symbols are used to stand for any things being talked about?

$a, b, c, \dots, x, y, \dots$

$a, b, c, \dots, x, y, \dots$

2. What can the things be that are talked about?

Numbers, like: 4, 8.57, -3 , $\frac{1}{2}$, $\sqrt{2}$, 3.14159, $\dots$

(A) Classes, like: "Horses, Animals, Cows, Mammals, $\dots$ " (B) The truth values (i.e., yes, no; or 1, 0) of statements such as: "Motor A is off." "Motor B is on." "Photocell C registers light."

3. What operations are there?

PLUS: $a + b$
MINUS: $a - b$
TIMES: $a \times b, ab$
DIVIDED BY: $a \div b, a/b$
ROOT: $\sqrt{a}$, etc.

OR: $a \vee b$
AND: $a \cdot b, ab$
NOT: $a', \neg a$
EXCEPT: $a \cdot b'$
OR ELSE: $a \wedge b$

4. What special constants are there?

ZERO, 0, such that $a + 0 = a$ for every a (and $a \cdot 0 = 0, a \neq \infty$)
ONE, 1, such that $a \cdot 1 = a$ for every $a, a \neq 0$
INFINITY, ∞ , such that $a + \infty = \infty$ for every a (and $a \cdot \infty = \infty$ for $a \neq 0$)

NULL CLASS, 0, such that $a \vee 0 = a$ for every a (and $a \cdot 0 = 0$)
UNIVERSE CLASS, 1, such that $a \vee 1 = 1$ for every a (and $a \cdot 1 = a$)

5. How many are all the things that are talked about?

INFINITY, ∞

2, or 4, or 8, or 16, or $\dots$, or ∞

6. What is an example of a rule?

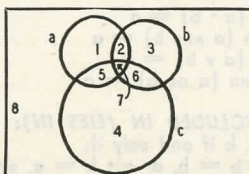
"The reciprocal of the reciprocal of a number is the number itself." $1/(1/a) = a$

"The truth value of the denial of the denial of a statement is the truth value of the statement itself." $(a')' = a$

7. How do you represent graphically the things talked about?

(A) By points on an infinite line:
See drawing on page 48
(B) By tables of numbers:

a	b
0	0
1	1
2	4
3	9
4	16
5	25
6	36



(A) By areas in a finite rectangle:
(See drawing below)
The areas No. 1 to 8 are:

No.	1: $ab'c'$	2: abc'	3: $a'b'c$	4: $a'b'c'$	5: $ab'c$	6: $a'bc$	7: abc	8: $a'b'c'$
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The null class has no location.
(B) By tables of truth values:

a	b	c	d
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

is beginning to be a change: Designers have begun to use the methods of Boolean algebra to effect the logical design of circuits to express behavior. For connecting the input and output yes's and no's, 1's and 0's, is a problem in Boolean algebra. What is Boolean algebra?

Boolean Algebra

Boolean algebra is a kind of algebra named after a great English mathematician, George Boole, who lived 1815 to 1864. He wrote a famous book called *The Laws of Thought*, in which he laid

out quite completely the design of a new algebra. It was somewhat like ordinary algebra but was adapted to the ideas and operations of logic, of reasoning.

Other mathematicians and symbolic logicians have since then considerably improved and extended the algebra which Boole devised.

Ideas of Boolean algebra

What are the ideas of Boolean algebra? In Chart 1 is a comparison of the main features of:

Elementary algebra, which we all

learn in school, for handling numbers, and which is essential for all computations in radio, electronics, electricity, etc., and

Boolean algebra, the newer algebra, which is useful for handling statements, classes, conditions, and circuit elements.

A great deal of information has been packed into this chart, and it is worth much attention.

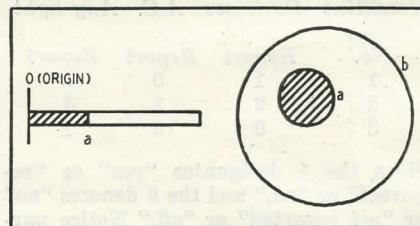
A reader may say: "There seem to be two ways in which Boolean algebra can be represented, by classes and by the truth values of statements." Yes—and there are more ways besides. Boolean algebra is an interesting mathematical framework that applies to quite a variety of different situations.

For example, take the number 30 and its factors 1, 2, 3, 5, 6, 10, 15, 30; let $a, b, c, \dots$ be any factors; let $a \vee b$ mean the least common multiple of a and b ; let $a \cdot b$ be the highest common factor of a and b ; and a' be $30/a$. You will find this to be a Boolean algebra.

For another example, consider sets of contacts, switches or relays; let $a, b, c, \dots$ be any switch contacts; $a = b$ if a is closed when b is closed and open when b is open; $a \vee b$ means a and b in parallel; $a \cdot b$ means a and b in series; and a' is any contact open when a is closed and closed when a is

CHART 2—THE RULES OF BOOLEAN ALGEBRA

TOPIC	ELEMENTARY ALGEBRA	BOOLEAN ALGEBRA
1. Commutative Law	$a + b = b + a$ a PLUS $b = b$ PLUS a $a \cdot b = b \cdot a$ a TIMES $b = b$ TIMES a	$a \vee b = b \vee a$ $(a \text{ OR } b) = (b \text{ OR } a)$ $a \cdot b = b \cdot a$ $(a \text{ AND } b) = (b \text{ AND } a)$
2. Associative Law	$(a + b) + c = a + (b + c)$ a PLUS b PLUS c is the same, whatever order you take them. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ a TIMES b TIMES c is the same, whatever order you take them.	$(a \vee b) \vee c = a \vee (b \vee c)$ a OR b OR c is the same, whatever order you take them. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ a AND b AND c is the same, whatever order you take them.
3. Distributive Law	ONE LAW: $a \cdot (b + c) = ab + ac$ a TIMES (b PLUS c) = (a TIMES b) PLUS (a TIMES c)	TWO LAWS: (1) $a \cdot (b \vee c) = a \cdot b \vee a \cdot c$ a AND (b OR c) = (a AND b) OR (a AND c) (2) $a \vee (b \cdot c) = (a \vee b) \cdot (a \vee c)$ a OR (b AND c) = (a OR b) AND (a OR c)
4. Combining	$a + a = 2a$ a PLUS $a = \text{TWO } a$ $a \cdot a = a^2$ a TIMES $a = a$ SQUARED There are numerical coefficients and exponents.	$a \vee a = a$ a OR $a = a$ $a \cdot a = a$ a AND $a = a$ There are no numerical coefficients or exponents.
5. Special Elements	0, ZERO; 1, UNITY $a + 0 = a$ $a \cdot 0 = 0$ $a + 1 = a + 1$ $a \cdot 1 = a$	0, NULL, NOTHING; 1, UNIVERSE, ALL $a \vee 0 = a$ (a OR NOTHING = a) $a \cdot 0 = 0$ (WHAT IS BOTH a AND NOTHING = NOTHING) $a \vee 1 = 1$ (a OR ALL = ALL) $a \cdot 1 = a$ (BOTH a AND ALL = a)
6. Opposites	TWO OPPOSITES: $a + (-a) = 0$ a PLUS (MINUS a) = 0 $a \times (1/a) = 1$ a TIMES (RECIPROCAL OF a) = 1 $-(-a) = a$ $1/(1/a) = a$	ONE OPPOSITE: $a \vee a' = 1$ a OR NOT $a = \text{ALL}$ $a \cdot a' = 0$ BOTH a AND NOT $a = \text{NOTHING}$ $(a')' = a$ (NOT-NOT- $a = a$) $1' = 0$ $0' = 1$
7. Laws Involving Opposites	$a + b = -(-a - b)$ $a \cdot b + a \cdot (-b) = 0$ $-(a + b + c \dots) = (-a) + (-b) + (-c) + \dots$	$a \vee b = (a' \cdot b')'$ a OR $b = \text{NOT (NOT-} a \text{ and NOT-} b)$ $a \cdot b = (a' \vee b')'$ a AND $b = \text{NOT (} a' \vee b' \text{)}$ $(a \vee b) \cdot (a \vee b') = a$ $(a \vee b \vee c \dots)' = a' \cdot b' \cdot c' \dots$
8. Absorption	$a + ab = a(1 + b)$ $a(a + b) = a^2 + ab$	$a \vee (a \cdot b) = a$ a OR (a AND b) = a $a \cdot (a \vee b) = a$ a AND (a OR b) = a
9. Relation of Less Than or Included In	LESS THAN OR EQUAL: $a \leq b$ if and only if $a + (\text{zero or some positive number}) = b$	INCLUDED IN (LIES IN): $a \subset b$ if and only if: $a \vee b = b$, or $a \cdot b = a$, or $a \cdot b' = 0$, or $a' \vee b = 1$



The drawings above indicate what a and b mean in standard and Boolean algebra.

open. This is typical Boolean algebra. How it is applied in practical electric circuits will be shown later.

Rules of Boolean algebra

But the ideas of Boolean algebra, interesting though they may be, are not enough: we also need the rules. We could work out the rules on the basis of ordinary reasoning. In fact, most ordinary reasoning is Boolean algebra, and is done by means of words and experience. But the rules of Boolean algebra expressed in mathematical form are potent and helpful. They are given in Chart 2, set into comparison with the rules of ordinary elementary algebra.

Here then is an introduction to the ideas and rules of Boolean algebra. Some of the ways to use them for dealing with circuit elements that can be "on" or "off," and some of the ways to use them for connecting input and output to express the behavior of a robot or mechanical brain, will be explained in the next article.

—end—